

INVESTIGATION OF THE HEAT CAPACITY OF $Y_{3-x}Eu_xSc_2Ga_3O_{12}$ ($0 < x < 3$) SOLID SOLUTIONS

Denisova L.T., Irtyugo L.A.

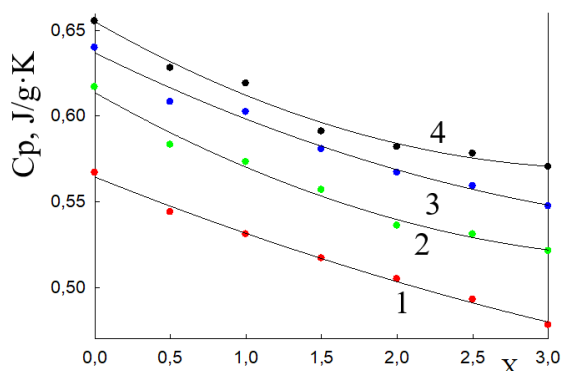
Siberian Federal University

660041, Krasnoyarsk, Svobodnyi pr. 79

Yttrium-scandium-gallium garnet is a promising material for high-efficiency phosphors. Similar ionic radii, identical charge, and close electronegativity of Eu^{3+} and Y^{3+} ions facilitate lattice doping, enabling Y^{3+} substitution and solid solution formation. Substituting Y^{3+} with Eu^{3+} has the potential to enhance the luminescent properties of this material. However, the thermodynamic and thermophysical properties of the $Y_{3-x}Eu_xSc_2Ga_3O_{12}$ system have not been systematically studied.

Solid solutions of the $Y_{3-x}Eu_xSc_2Ga_3O_{12}$ system ($0 < x < 3$) were synthesized by a high-temperature solid-state reaction at 1793 K using B_2O_3 as a flux. The phase purity of the obtained samples was verified by XRD using a Bruker D8 diffractometer with a VANTEC -1 linear detector and $CuK\alpha$ radiation. The synthesized garnets belong to the cubic space group $1a\bar{3}d$.

The temperature dependence of the heat capacity for all obtained substances was determined by differential scanning calorimetry (DSC) using a STA 449 S Jupiter instrument. The experiments were conducted in platinum crucibles with lids, with a heating rate of 20 K/min, from 323 K to 1023 K in an air atmosphere. The obtained results are presented in the figure.



Dependence of the specific heat capacity on the composition of $Y_{3-x}Eu_xSc_2Ga_3O_{12}$:
1 – $T = 323$ K, 2 – $T = 523$ K, 3 – $T = 723$ K, 4 – $T = 923$ K

Substitution of Y^{3+} ions with Eu^{3+} ions leads to a nonlinear decrease in heat capacity in the $Y_{3-x}Eu_xSc_2Ga_3O_{12}$ system throughout the studied temperature range.

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